

B.TECH/ M.Tech (Integrated) DEGREE EXAMINATION, MAY 2025

Fourth Semester

21CSE255T - COMPUTER GRAPHICS AND ANIMATION*(For the candidates admitted during the academic year 2021-2022 to 2024-2025)***Note:**

- i. **Part - A** should be answered in OMR sheet within first 40 minutes and OMR sheet should be handed over to hall invigilator at the end of 40th minute.
 ii. **Part - B** and **Part - C** should be answered in answer booklet.

Time: 3 hours**Max. Marks: 75****PART - A (20 × 1 = 20 Marks)**

Answer all Questions

Marks BL CO

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|--|---|---|---|
| 1. The video device with reduced volume, power consumption, and weight is
(A) CRT
(B) Flat-panel display
(C) Portable display
(D) Color CRT | 1 | 1 | 1 |
| 2. _____ is a technique that is used to approximate the halftones without the reduction of spatial resolution.
(A) Halftoning
(B) Dithering
(C) Error diffusion
(D) Interpolation | 1 | 2 | 1 |
| 3. _____ is responsible for accessing the frame buffer to refresh the screen.
(A) Graphics package
(B) CPU
(C) Video controller
(D) Monitor | 1 | 2 | 1 |
| 4. On a monochromatic monitor, the frame buffer is known as
(A) Display file
(B) Pixmap
(C) Bitmap
(D) Refresh buffer | 1 | 2 | 1 |
| 5. We rotate a point P (2, 5) by 60° about a pivot point (1, 2). The new transformed point P' would be:
(A) (-4, 1)
(B) (1, -4)
(C) (-1, 4)
(D) (1, 4) | 1 | 3 | 2 |
| 6. What will be the new position points after 2D translation is performed for the points (15,18) with the translation distance (7,8)
(A) (25,26)
(B) (22,26)
(C) (22,28)
(D) (26,26) | 1 | 3 | 2 |
| 7. Which of the following statements about reflection transformations is true?
(A) Reflection preserves angles but not distances.
(B) Reflection does not change the orientation of an object.
(C) Reflection can only be performed about the x-axis or y-axis.
(D) Reflection results in a change of the object's area. | 1 | 2 | 2 |
| 8. When the line is parallel to the boundaries then what is the value of p_k ?
(A) $p_k < 0$
(B) $p_k > 0$
(C) $p_k = 0$
(D) $p_k = 1$ | 1 | 2 | 2 |
| 9. In 3D viewing we have
(A) 3D window and 3D viewport
(B) 3D window and 2D viewport
(C) 2D window and 2D viewport
(D) 2D window and 3D viewport | 1 | 2 | 3 |
| 10. Which rendering method traces the path of light rays from the eye to determine visibility and lighting?
(A) Back-Face Detection
(B) Scan-Line Method
(C) Ray-Tracing Method
(D) Octree Method | 1 | 2 | 3 |

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| 11. _____ is a characteristic of a perspective projection. | 1 | 2 | 3 |
| (A) Parallel lines remain parallel | | | |
| (B) Objects farther from the viewer appear smaller | | | |
| (C) The projection plane is parallel to the view plane | | | |
| (D) It preserves the object dimensions accurately | | | |
| 12. In 3D viewing, the function of the viewport transformation is _____. | 1 | 1 | 3 |
| (A) To map the clipping window to the viewport | | | |
| (B) To perform perspective division | | | |
| (C) To clip objects outside the view volume | | | |
| (D) To normalize the coordinates | | | |
| 13. In the scan-line rendering method, which value is used to determine the visibility of a surface? | 1 | 2 | 4 |
| (A) The normal vector of the surface | | | |
| (B) The depth value of the surface at each pixel | | | |
| (C) The reflectance property of the material | | | |
| (D) The texture mapping coordinates | | | |
| 14. The visible-surface detection algorithm that works by recursively subdividing a scene until each area corresponds to a single visible surface is _____. | 1 | 2 | 4 |
| (A) Depth-buffer method | | | |
| (B) Scan-line method | | | |
| (C) Area-subdivision method | | | |
| (D) Ray-casting method | | | |
| 15. The technique often used to improve the visual quality of low-resolution images by displaying light intensities is _____. | 1 | 2 | 4 |
| (A) BSP-Tree Method | | | |
| (B) Depth-Sorting Method | | | |
| (C) Dithering Techniques | | | |
| (D) Curved Surface Approximation | | | |
| 16. The image format known for supporting transparency and lossless compression, making it ideal for graphics with sharp edges, is _____. | 1 | 2 | 4 |
| (A) JPEG | | | |
| (B) PNG | | | |
| (C) GIF | | | |
| (D) BMP | | | |
| 17. The main purpose of converting between different color models is _____. | 1 | 1 | 5 |
| (A) To enhance image resolution | | | |
| (B) To adjust contrast and brightness | | | |
| (C) To adapt colors for different devices and applications | | | |
| (D) To change an image's shape | | | |
| 18. The technique is commonly used to improve the contrast of an image by redistributing pixel intensity values | 1 | 2 | 5 |
| (A) Morphological Filtering | | | |
| (B) Image Compression | | | |
| (C) Edge detection | | | |
| (D) Histogram Equalization | | | |
| 19. An example of a contrast stretching transformation function is _____. | 1 | 2 | 5 |
| (A) Median filtering | | | |
| (B) Discrete Fourier Transform | | | |
| (C) Linear transformation | | | |
| (D) Huffman encoding | | | |
| 20. In contrast stretching, what happens to pixel values in a low-contrast image? | 1 | 2 | 5 |
| (A) They are compressed into a narrow range | | | |
| (B) They are replaced with random values | | | |
| (C) They are stretched over the entire intensity range | | | |
| (D) They remain unchanged | | | |

PART - B (5 × 8 = 40 Marks)

Answer all Questions

Marks BL CO

21. (a) Explain the operational mechanism behind the Direct View storage tube and how it functions to display images with precision and reliability.

(OR)

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|--|---|---|---|
| (b) Explain the concept of pixel addressing and discuss different filled-area primitives with suitable algorithms. | 8 | 2 | 1 |
| 22. (a) A company wants to develop a graphics application where users can zoom in and out on a 2D image. Explain how scaling transformation can be applied to achieve this effect. Provide the necessary transformation matrix and an example calculation. | 8 | 3 | 2 |
| (OR) | | | |
| (b) A CAD software allows users to reflect a shape across a specified axis. Given a triangle with vertices A(2,3), B(5,6), and C(7,2), apply reflection across the X-axis and determine the new coordinates. | | | |
| 23. (a) As a game developer designing a 3D world for a 2D screen, you need to use projection techniques to represent depth and perspective accurately. What is projection, what are its different types, and how does Cavalier projection work with an example? | 8 | 3 | 3 |
| (OR) | | | |
| (b) Imagine you are an architect designing a realistic 3D model of a building and need to create perspective drawings to represent depth accurately. How does the concept of a vanishing point help in achieving this, and how does it function in different types of perspective projections? | | | |
| 24. (a) How does an illumination model calculate light intensity for ambient and diffuse reflection components on different surfaces within a scene? | 8 | 2 | 4 |
| (OR) | | | |
| (b) What role does dithering play in computer graphics, and how does it improve the visual quality of low-bit-depth images? | | | |
| 25. (a) How are motion specifications defined in computer graphics, and what methods are used to control movement within animations? | 8 | 2 | 5 |
| (OR) | | | |
| (b) Elaborate on how the RGB and CMY color models depict colors in digital images and graphics through a different perspective. | | | |

PART - C (1 × 15 = 15 Marks)

Answer any 1 Questions

Marks BL CO

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| 26. Imagine you are developing a computer graphics application that requires rendering a circle. Given the center coordinates (4, -4) and a radius of 10, explain how you would generate and display all the points forming this circle across all quadrants using the Midpoint Circle Algorithm. Describe the algorithm's steps and how it ensures accurate representation in your application. | 15 | 4 | 1 |
| 27. In a navigation system for a drone flying over a defined region, you need to ensure the drone stays within a rectangular window. The drone's path is represented by a line segment with endpoints P1(10,30) and P2(80,90), while the window boundaries are defined by the points A(20,20), B(90,20), C(90,70), and D(20,70). Your task is to develop an algorithm to clip the drone's path against the window boundaries using the Cohen-Sutherland line clipping algorithm. This ensures that the drone remains within the specified region. Can you guide us through the steps of the algorithm, determining which portions of the path lie inside or outside the window, and adjusting the endpoints accordingly? | 15 | 4 | 2 |

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THE HELPERS